Automated Knowledge Extraction from Scientific Literature

Sophia Chen, David K. Lin, Elena Rostova

MIT Computer Science and Artificial Intelligence Laboratory; ETH Zurich

Published: 2024 | DOI: 10.1109/TKDE.2024.3391024

Abstract

Extracting structured knowledge graphs from unstructured scientific publications accelerates meta-analysis and scholarly discovery. We present SciExtract, a pipeline leveraging transformer-based language models fine-tuned on computer science and biomedical corpora. SciExtract achieves high precision in entity linking, relation classification, and citation resolution, facilitating automatic construction of cross-disciplinary literature graphs.

Keywords: Knowledge Graphs, Natural Language Processing, Transformers, Information Extraction, Machine Learning

1. Introduction

The rate of global academic publication exceeds three million papers annually, presenting severe information overload for researchers. Traditional bibliometric indexes only capture co-authorship and raw citations without extracting granular semantic relations between claims, methods, and outcomes.

2. The SciExtract Architecture

SciExtract employs a two-tier extraction pipeline. First, specialized layout-aware parsers extract sections and bibliographies. Second, fine-tuned transformer encoders identify scientific entities (tasks, datasets, metrics) and classify relational links.

3. Cross-Paper Knowledge Synthesis

By linking extracted entities into an interconnected knowledge graph, SciExtract enables multi-hop reasoning across independent publications, surfacing hidden thematic bridges between disparate research sub-fields.

References

[1] Vance, A., Rostova, E., and Thorne, M. "Attention-Based Graph Neural Networks for Knowledge Discovery." ACM Transactions, 2024.

[2] Beltagy, I., Lo, K., and Cohan, A. "SciBERT: A Pretrained Language Model for Scientific Text." EMNLP, 2019.

[3] Rostova, E., and Vance, A. "Scalable Semantic Search Using Vector Embeddings." Journal of Machine Learning Research, 2023.